# Formulating MDP

1. States (S) :- Fresh ($S_1$), Stress ($S_2$), Tired ($S_3$)
2. Actions (A) :- Study ($A_1$), Sleep ($A_2$), Hang Out ($A_3$)
3. Transitions Probability (P) :-

| State (S) | Action (A) | Fresh ($S_1$) | Stress ($S_2$) | ~~Hang out~~ Tired ($S_3$) |
|---|---|---|---|---|
| Fresh ($S_1$) | $A_1$ (Study) | 0.20 | 0.10 | 0.70 |
| | $A_2$ (Sleep) | 0.80 | 0.10 | 0.10 |
| | $A_3$ (Hang out) | 0.30 | 0.10 | 0.60 |
| Stress ($S_2$) | $A_1$ | 0.20 | 0.50 | 0.30 |
| | $A_2$ | 0.80 | 0.10 | 0.10 |
| | $A_3$ | 0.70 | 0.10 | 0.20 |
| Tired ($S_3$) | $A_1$ | 0.10 | 0.50 | 0.40 |
| | $A_2$ | 0.90 | 0.10 | 0.00 |
| | $A_3$ | 0.30 | 0.20 | 0.50 |

4. Reward (R) :-

| State (S) ↓ / Action → | Study ($A_1$) | Sleep ($A_2$) | Hang out ($A_3$) |
|---|---|---|---|
| Fresh ($S_1$) | +10 | +2 | −5 |
| Stress ($S_2$) | +6 | +8 | +4 |
| Tired ($S_3$) | −1 | +8 | +1 |

5.> Discount Factor ($\gamma$) = 0.9

Formula for state value V(S) :-

$$V_{k+1}(S) = \max_{a} \left\{ \sum P(s' | s, a) \left[ R(s, a, s') + \gamma V_k(s') \right] \right\}.$$

Initially,

$V_0(S_1) = V_0(S_2) = V_0(S_3) = 0$

## Step 1 :- Computing $V_1(S)$

| State(S) | Action(A) | Calculation | | Best Action. | $V_1(S)$ |
|---|---|---|---|---|---|
| Fresh | Study | 0.2 (10) + 0.1 (10) + 0.7 (10) | 10 | Study | 10 |
| | Sleep | 0.8 ( 2) + 0.1 (2) + 0.1 (+2) | 2 | | |
| | Hang out | 0.3(−5) + 0.1 (−5) + 0.6(−5) | −5 | | |
| Stress | Study | 0.2(6) + 0.5 (6) + 0.3(6) | 6 | Sleep | 8 |
| | Sleep | 0.8(8) + 0.1 (8) + 0.1 (8) | 8 | | |
| | Hang out | 0.7(4) + 0.1 (4) + 0.2 (4) | 4 | | |
| Tired. | Study | 0.1(−1) + 0.5(−1) + 0.4 (−1) | −1 | Sleep | 8 |
| | Sleep | 0.9 (8) + 0.1(8) + 0 (8) | 8 | | |
| | Hang out | 0.3 (1) + 0.2(1) + 0.5(1) | 1 | | |

## Step 2 :- Computing $V_2(S)$

| State | Action(A) | |
|---|---|---|
| Fresh | Study | 0.2[(10) + 0.9 (10)] + 0.1[(10) + 0.9 (8)] + 0.7[(10) + 0.9 (8)] = 17.56 |
| | Sleep | 0.8 [(2) + 0.9(10)] + 0.1[(2) + 0.9(8)] + 0.1[(2) + 0.9(8)] = 10.64 |
| | Hang out | 0.3[(−5) + 0.9 (10)] + 0.1 [(−5) + 0.9 (8)] + 0.6 [(−5) + 0.9(8)] = 2.74 |
| Stress | Study | 0.2[6 + 0.9(10)] + 0.5[6 + 0.9(8)] + 0.3[6 + 0.9(8)] = 13.56 |
| | Sleep | 0.8 [8 + 0.9 (10)] + 0.1 [8 + 0.9 (8)] + 0.1[8 + 0.9(8)] = 16.64 |
| | Hang out | 0.7 [4 + 0.9 (10)] + 0.1 [4 + 0.9(8)] + 0.2[4 + 0.9(8)] = 12.46 |
| Tired | Study | 0.1[(−1) + 0.9 (10)] + 0.5[(−1) + 0.9(8)] + 0.4 [(−1) + 0.9(8)] = 6.38 |
| | Sleep | 0.9 [8 + 0.9 (10)] + 0.1 [8 + 0.9(8)] + 0 [8 + 0.9 (8)] = 16.82 |
| | Hang out | 0.3[1 + 0.9(10)] + 0.2[1 + 0.9(8)] + 0.5[1 + 0.9(8)] = 8.74 |

$V_2(S_1) = 17.56$

$V_2(S_2) = 16.64$

$V_2(S_3) = 16.82$